

The Backbone Clue



Asha and Ben hurried into the sea life center to escape the rain.

Dr. Marina met them beside the touch tank.

A silver fish flashed past the glass, while a jellyfish floated like a glowing umbrella.

How can they both be animals if their bodies look so different? Ben asked.

Dr. Marina smiled. Look for one important clue: a backbone.



Meet the Vertebrates



Dr. Marina pointed to a fish skeleton model.

Some animals have a backbone inside the body. They are called vertebrates.

A backbone is made of small bones called vertebrae.

Fish, amphibians, reptiles, birds, and mammals are vertebrates.

So a turtle and a fish belong in the same big group, Asha said.

Yes, said Dr. Marina. Different shapes, same clue.



The Invertebrate Puzzle



At the touch tank, Dr. Marina gently showed them a sea star.

This animal has no backbone, she said. Animals without backbones are called invertebrates.

Ben watched a jellyfish pulse through the water.

How do they move without one?

They have other body plans, said Dr. Marina. Some are soft, some have shells, and some have hard outer coverings.



Backbone Check



The children became animal detectives.

Asha placed fish and turtle cards under vertebrates.

Ben placed jellyfish, sea star, crab, and octopus under invertebrates.

The clue is not where the animal lives, Dr. Marina said.

The clue is what is inside its body.

Ben grinned. A backbone is the secret clue!



Mystery Solved



At the last tank, a colorful fish swam past a floating jellyfish. Asha looked from one animal to the other. One has a backbone, and one does not, she said. Exactly, said Dr. Marina. Both are important. They are just built in different ways. The rain had stopped outside, but the mystery was already solved.



Science Element



Vertebrates are animals with backbones inside their bodies. Examples include fish, amphibians, reptiles, birds, and mammals. Invertebrates are animals without backbones.

Examples include jellyfish, worms, insects, spiders, octopuses, crabs, and sea stars.

Simple idea: some animals have a backbone inside, and some do not.

Scientists use this clue to group animals and understand how they live.

